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DRISHTI Innovation Challenge

AURORA

Automated Unified Reasoning and Optimization Architecture

Proposed Solution · Annexure 1

Problem Statement-71

Field	Details
Applicant Name	Leejo Purvathan
Company Name	Arcaisys
Challenge	Problem Statement-71 , Design Optimization, Compliance Validation & CAD-BOM Synchronization
Duration	20 Months
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1. Brief Summary of the Proposed Solution

Naval ship design workflows currently rely on manual cross-verification of CAD models, BOM data, and classification rules, leading to long validation cycles, high error risk, and operational inefficiencies. The proposed solution, AURORA (Automated Unified Reasoning and Optimization Architecture), is a defence-grade decision-support system that unifies design data, automates compliance validation, and synchronises CAD and BOM information within a single on-premise environment. It enables engineers to rapidly analyse design changes, detect inconsistencies, and evaluate impacts across complex, multi-component ship systems.

Problem Statement-71 addresses Design Optimization, Compliance Validation, and CAD-BOM Synchronization: engineers presently cross-check drawings, PLM records, and classification rule libraries through review cycles spanning weeks, making the process error-prone and operationally expensive.

OVERVIEW OF THE PROPOSED SOLUTION

AURORA is proposed as an integrated, on-premise decision-support system that transforms the existing fragmented and manual naval design validation workflow into a unified, automated, and traceable process. It addresses design optimization, compliance validation, and CAD-BOM synchronization within a single platform.

Unified Design Data Environment

The system establishes a single source of truth by creating a unified digital representation of the ship design. CAD data, Bill of Materials (BOM), classification rules, and simulation outputs are continuously synchronized, eliminating inconsistencies that currently require manual cross-verification.

Automated Data Ingestion and Mapping

AURORA ingests multi-format CAD models, BOM data, and classification rule libraries into a centralized environment. Each component is digitally mapped and linked across datasets, enabling accurate alignment between design, production, and compliance information.

CAD-BOM Synchronization and Discrepancy Detection

The system continuously reconciles CAD assemblies with BOM records to identify mismatches such as missing components, material inconsistencies, mass deviations, and revision conflicts. All discrepancies are clearly flagged for rapid resolution, ensuring data consistency across the lifecycle.

Automated Compliance Validation

AURORA performs automated validation of design parameters against multiple classification society rules. It generates auditable, rule-level findings and ensures that compliance checks are consistent, traceable, and significantly faster than manual processes.

Change Impact Analysis

Any design modification triggers a structured impact analysis that identifies all dependent components and systems. This enables engineers to understand downstream effects before implementation, preventing cascading issues and reducing rework.

Change-Driven Validation Workflow

Instead of re-evaluating the entire system, AURORA focuses validation only on components affected by a design change. This targeted approach significantly reduces validation time while maintaining accuracy and completeness.

Design Decision Support

The system provides advisory design alternatives where trade-offs exist between parameters such as weight,



cost, and compliance. These recommendations support informed decision-making while ensuring that final authority remains with engineers.

Integrated Dashboard and Reporting

All outputs—including compliance results, discrepancy reports, impact analyses, and design recommendations—are presented through a unified dashboard. Engineers and programme managers can interact with results, generate reports, and track system status in real time.

Auditability and Human Oversight

AURORA maintains a complete audit trail of all system actions, validations, and approvals. All outputs are explainable and subject to engineer review, ensuring transparency and alignment with defence-grade operational requirements.

Secure On-Premise Deployment

The entire system is deployed in an air-gapped, on-premise environment with no dependency on external connectivity. This ensures compliance with strict security and data protection requirements.

Key Features & Innovation

AURORA introduces a unified Digital Twin and dependency-driven system model to represent all ship components and their interrelationships. It combines deterministic rule-based validation for compliance-critical tasks with selective AI-based modules for pattern detection and optimisation. The system supports multi-society compliance evaluation, automated CAD-BOM discrepancy detection, and generation of optimised design alternatives with trade-off analysis. All outputs are explainable, traceable, and require engineer validation, ensuring reliability in defence-grade applications.

Overall Impact and Value Proposition

The system will be deployed as an air-gapped, on-premise platform integrated with existing CAD, PLM, and ERP systems. It streamlines the end-to-end workflow from design change input to compliance validation and reporting. Expected outcomes include reducing validation timelines from weeks to hours, improving accuracy in discrepancy detection, and enabling informed design decisions through structured impact analysis and optimisation support.

2. Key Technologies Used

#	Technology	Role
1	Neo4j Graph Database	Core knowledge representation , directed dependency graph with weighted edges; BFS-based impact propagation across 10,000+ component nodes
2	OpenCascade Technology (OCC)	Multi-format CAD geometry parsing, feature extraction, shape healing, and assembly decomposition (STEP/IGES/SolidWorks/CATIA)
3	XGBoost / LightGBM	CAD-BOM matching confidence scoring with SHAP explainability



#	Technology	Role
4	NSGA-II Genetic Algorithm (pymoo)	Optional design optimisation module: weight, cost, and compliance trade-off analysis with Pareto front output. Applied selectively where design parameter space warrants it.
5	Multi-Society Rule Engine	Deterministic compliance evaluation against IRS, Lloyd's Register, DNV, Bureau Veritas, and ABS rule catalogs

3. Deliverables

S.No	Deliverable Name	Type	Brief Description
D-01	Digital Twin Core	Data + Engine	Unified parametric model integrating CAD geometry, BOM metadata, simulation data, and rule bindings for every ship component.
D-02	Dependency Knowledge Graph	Database (Neo4j)	Directed weighted graph encoding all inter-system dependencies with documented relationship taxonomy.
D-03	Impact Propagation Engine	Software Module	BFS/DFS traversal with exponential depth decay; produces ranked impact lists with propagation path explanations.
D-04	Decision Output Engine	Software Module	Consolidated engineer-facing report: compliance status, BOM sync errors, and optimized design recommendations.
D-05	System Integration Package	APIs + Docs	REST API layer, PLM adapter stubs, CAD system webhooks, and full OpenAPI 3.0 specification.
D-06	Security & Deployment Package	Infrastructure	Docker-Compose/Kubernetes manifests for air-gapped on-premise deployment.
D-07	Validation & Acceptance Report	Document	End-to-end system validation against real engineering scenarios; accuracy metrics; SME sign-off certificate.
D-08	CAD Parsing & Normalisation Engine	Software Module	Extracts geometry features, assembly hierarchy, mass properties from STEP, IGES, SolidWorks using PythonOCC.
D-09	CAD-BOM Synchronisation Engine	Software Module	Multi-method matching (geometry hash, name similarity, weight delta) with XGBoost confidence scoring; structured discrepancy reports.



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D-10	Multi-Society Rule Compliance Engine	Software Module	Deterministic rule evaluation against IRS, Lloyd's Register, DNV, Bureau Veritas, and ABS with per-rule audit trail. Operates in two modes: (a) full baseline compliance scan; (b) change-triggered delta mode, where only rules affected by the specific attribute changes in a drawing revision are re-evaluated, with findings explicitly linked to the drawing change that triggered them.
D-11	Design Optimisation Engine (NSGA-II)	Software Module	NSGA-II multi-objective optimisation: weight, cost, and compliance objectives; Pareto front output; validated against FEA. Activation requires ≥ 3 quantifiable design parameters with defined bounds; if insufficient parameters are available for a given ship project, the module delivers a documented parameter sensitivity analysis in lieu of Pareto optimisation.
D-12	AURORA Decision Intelligence Dashboard	Software Module	The AURORA Decision Intelligence Dashboard is a React + TypeScript on-premise web application providing engineers and programme managers a unified interface across all engine outputs.
D-13	Drawing Synchronisation & Revision Engine	Software Module	Automated propagation of approved design changes into production drawings. Handles DXF/DWG and PDF drawing formats via ezdxf (DXF) and reportlab/pypdf (PDF). Drawing-CAD mismatches (geometry hash mismatch between drawing and current CAD state) are flagged as DRAWING_STALE discrepancies. Targets PLM drawing vaults (Teamcenter/Windchill) via adapter stubs; fallback is file-system folder watch on a designated drawing directory.
D-14	Validation Benchmarking Report	Document	Structured accuracy report covering: CAD-BOM mismatch detection (Precision $\geq 90\%$, Recall $\geq 90\%$); compliance validation agreement with SME $\geq 85\%$; zero critical false negatives in compliance checks; false positive/negative analysis; SME sign-off documentation.
D-15	Classification Society Submission Package	Software Module + Document	Generates structured approval submission packages for IRS/LR/DNV/BV/ABS: compliance summary report (PDF, rule-by-rule findings), design calculation sheets, drawing revision log, and a discrepancy resolution record. Includes a Society Query Ingestion workflow: society



S.No	Deliverable Name	Type	Brief Description
			comments imported as structured change requests (JSON), automatically linked to affected components in the knowledge graph, triggering targeted re-evaluation and revised submission generation. Supports the full approval cycle: Pre-submission check → Submit → Receive queries → Resolve → Re-submit.

4. Proposed Timeline

Phase	Month(s)	Key Activities	Milestone
Phase 0 , Foundation	1–2	Stakeholder interviews; data audit; classification rule catalog curation; IT infrastructure assessment; dev environment setup.	Signed-off data schema and approved rule catalog
Phase 1 , Core Engines	3–5	Knowledge graph construction; CAD parsing engine; Digital Twin data model; multi-society compliance engine v1; basic CAD-BOM synchronisation.	All core AURORA engines operational in dev
Phase 2 , ML & Optimization	6–7	XGBoost CAD-BOM matching model; NSGA-II optimisation engine; SHAP explainability layer; end-to-end pipeline integration.	Integrated ML pipeline with full end-to-end outputs
Phase 3 , Dashboard & Integration	8–9	Decision Intelligence Dashboard UI; RBAC implementation; PDF report engine; REST API gateway; PLM adapter stubs; security hardening. Classification Society Submission Package : report formatter, society query ingestion API, re-submission workflow.	Secured on-premise dashboard deployed
Phase 4 , Pilot & Validation	10–12	Pilot execution: 4 historical scenarios; CAD-BOM matching accuracy (Precision $\geq 90\%$, Recall $\geq 90\%$); compliance validation agreement with SME $\geq 85\%$; zero critical false negatives in compliance checks; UAT sign-off.	Validated accuracy vs. historical actuals; UAT sign-off
Phase 5 , Handover & Docs	13–14	Final model retraining; system documentation; 2-day user training workshop; Field Operations Manual; source code handover.	Full system handover with acceptance certificate
Phase 6 , Production Enhancements	15–20	Stability improvements; integration refinement with MDL PLM/ERP ecosystem	Production-grade enterprise system deployed



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5. Project Milestones

M#	Milestone Name	Month	Completion Criteria	Deliverables	Phase Total (₹)
M-01	Foundation Complete	2	Data schema signed off; rule catalog curated; dev environment operational.	Schema Docs, Rule Catalog	₹20,50,000
M-02	Core Engines Ready	5	All core engines (graph, CAD parser, compliance engine, Digital Twin) pass unit and integration tests.	D-01 to D-03, D-08	₹29,00,000
M-03	ML Pipeline Integrated	7	ML models trained and validated; full end-to-end pipeline from input to output functional.	D-09 to D-11	₹20,50,000
M-04	Dashboard Deployed	9	Secured on-premise dashboard with RBAC; PDF report generation functional; security audit passed.	D-04 to D-06, D-13,D-14	₹18,50,000
M-05	Pilot Validation Passed	12	All pilot scenarios executed; CAD-BOM match ≥90%; compliance accuracy benchmarks met; UAT sign-off.	Pilot Report, UAT Sign-off	₹25,00,000
M-06	System Handover	14	Full source code delivered; documentation complete; 2-day training conducted; acceptance report signed.	D-07, FOM, Training Report	₹13,50,000
M-07	Production System Live	20	All Phase 6 enhancements deployed; integration refinement complete; expanded rule library validated; final acceptance certificate.	Production Deployment Report	₹53,00,000
Total Project Cost					₹1,80,00,000

***This estimate is based on current project requirements and is subject to change upon any scope modifications, unforeseen technical complexities, or third-party fee adjustments with the requirement that funds for each milestone must be released within 2-3 weeks of Milestone completion.**



6. Abbreviations & Technical Glossary

The following table defines all abbreviations and technical terms used throughout this document.

Abbreviation / Term	Full Form / Explanation
ABS	American Bureau of Shipping , one of the five international classification societies whose rules AURORA validates against.
AI	Artificial Intelligence , the broader field enabling AURORA's intelligent design optimisation, compliance validation, and BOM synchronisation.
API	Application Programming Interface , a defined contract by which software modules communicate; AURORA exposes all capabilities via REST APIs.
AURORA	Automated Unified Reasoning and Optimization Architecture , Arcaisys's proposed system for Problem Statement-71 addressing design optimisation, compliance validation, and CAD-BOM synchronisation.
BFS	Breadth-First Search , the graph traversal algorithm used by the Impact Propagation Engine to discover all components affected by a design or compliance change.
BOM	Bill of Materials , the structured list of all components, sub-assemblies, and raw materials constituting a ship system; AURORA synchronises this against the CAD model.
BREP	Boundary Representation , a method for representing shapes in 3D CAD using surfaces, edges, and vertices; the internal geometry format used by OpenCascade Technology (OCC).
BV	Bureau Veritas , French international classification society; one of the five rule sources in the AURORA multi-society compliance engine.
CAD	Computer-Aided Design , software used to create and modify digital 3D models of ship components and assemblies; AURORA ingests files from CATIA, NX, SolidWorks, and others.
CAD-BOM Sync	The process of automatically matching and reconciling each CAD assembly node to its corresponding BOM line item to detect discrepancies in mass, material, or quantity.
CATIA	Computer-Aided Three-dimensional Interactive Application , a widely used CAD software by Dassault Systèmes; one of the formats supported by AURORA's CAD parsing engine.
CFD	Computational Fluid Dynamics , numerical simulation of fluid flow around ship structures; AURORA's Simulation Integration Layer ingests CFD results and binds them to the Digital Twin.



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Abbreviation / Term	Full Form / Explanation
Clash Detection	A Phase 6 capability that automatically identifies geometric interferences (collisions) between components in the 3D ship model before physical construction.
DAG	Directed Acyclic Graph , a graph structure with directed edges and no cycles; used by AURORA to represent system dependency hierarchies and impact propagation paths.
DFS	Depth-First Search , an alternative graph traversal strategy used alongside BFS in the impact propagation engine for deep dependency chain analysis.
DNV	Det Norske Veritas , Norwegian international classification society; one of the five rule sources in the AURORA multi-society compliance engine.
Docker / Kubernetes	Containerisation and orchestration technologies used to package and deploy AURORA's microservices in an air-gapped on-premise environment.
ERP	Enterprise Resource Planning , business management software (e.g., SAP, Oracle) integrated with AURORA for BOM synchronisation and cost rate table ingestion.
FEA	Finite Element Analysis , numerical simulation technique for computing structural stress, deflection, and fatigue in ship components; AURORA's surrogate model approximates FEA outputs.
FOM	Field Operations Manual , the operational documentation delivered to the end-user organisation enabling independent operation and maintenance of AURORA.
GPU	Graphics Processing Unit , hardware accelerator used to train ML models (PyTorch surrogate, XGBoost) and run NSGA-II optimisation significantly faster than CPU-only compute.
IACS	International Association of Classification Societies , the umbrella body coordinating standards across member societies including IRS, DNV, Lloyd's Register, BV, and ABS.
IGES	Initial Graphics Exchange Specification , a neutral CAD file format; one of the formats supported by AURORA's CAD parsing engine via OCC.
IRS	Indian Register of Shipping , India's national ship classification society; the primary rule source for AURORA's compliance engine.
JT	Jupiter Tessellation , a lightweight 3D CAD data format used for visualisation and collaboration; supported by AURORA's OCC-based CAD parser.
JSON	JavaScript Object Notation , a lightweight data interchange format used by AURORA's internal APIs to pass structured data between modules.
LDAP / AD	Lightweight Directory Access Protocol / Active Directory , enterprise directory services used by AURORA's RBAC module for user authentication on-premise.
Levenshtein	An edit-distance metric measuring the minimum number of character-level edits required to transform one string into another; used by AURORA's CAD-BOM name similarity signal.



Abbreviation / Term	Full Form / Explanation
LightGBM	Light Gradient Boosting Machine , a high-performance ML framework used alongside XGBoost for CAD-BOM confidence scoring, offering faster training on large datasets.
LR	Lloyd's Register , British international classification society; one of the five rule sources in the AURORA multi-society compliance engine.
MinIO	An open-source, S3-compatible object storage system used within AURORA's on-premise deployment for storing CAD files, simulation results, and model artifacts.
ML	Machine Learning , a sub-field of AI; AURORA uses ML for CAD-BOM matching confidence scoring (XGBoost) and FEA surrogate modelling (PyTorch).
mTLS	Mutual Transport Layer Security , a security protocol requiring both client and server to authenticate each other; used in AURORA's Phase 6 zero-trust architecture.
Neo4j	A leading graph database management system; used as the storage and query engine for AURORA's Dependency Knowledge Graph.
NLP	Natural Language Processing , a branch of AI enabling computers to understand human language; used in AURORA for rule text parsing and discrepancy narrative generation.
NSGA-II	Non-dominated Sorting Genetic Algorithm II , a multi-objective evolutionary optimisation algorithm implemented via the pymoo library; AURORA uses it to simultaneously minimise weight, cost, and maximise compliance.
OCC / PythonOCC	OpenCascade Technology , an open-source 3D CAD kernel; PythonOCC provides Python bindings. AURORA uses OCC for multi-format CAD parsing, shape healing, feature extraction, and geometry hashing.
OpenAPI 3.0	A widely adopted specification standard for describing REST APIs; AURORA's system integration package includes a full OpenAPI 3.0 specification for all exposed endpoints.
Pareto Front	The set of optimal solutions to a multi-objective optimisation problem where no single objective can be improved without degrading another; AURORA's NSGA-II engine outputs the Pareto front for design decision support.
PKI	Public Key Infrastructure , a cryptographic framework used for digital signatures in AURORA's change approval workflows.
PLM	Product Lifecycle Management , software systems (e.g., Siemens Teamcenter, PTC Windchill) managing engineering data across a product's lifecycle; AURORA provides PLM adapter stubs.
pymoo	A Python-based multi-objective optimisation library; AURORA uses it to implement the NSGA-II genetic algorithm for design optimisation.



Abbreviation / Term	Full Form / Explanation
PyTorch	An open-source ML framework; AURORA uses PyTorch to train the FEA surrogate neural network model that approximates structural simulation outputs from design parameters.
RBAC	Role-Based Access Control , a security model restricting system access based on user roles (Viewer, Analyst, Approver, Admin) as implemented in AURORA's dashboard.
REST	Representational State Transfer , an architectural style for designing networked APIs using standard HTTP methods; all AURORA interfaces are REST-based.
SAP / Oracle	Major ERP software vendors; AURORA's integration layer includes connectors for SAP PM and Oracle for BOM and cost data bi-directional synchronisation.
SHA-256	Secure Hash Algorithm 256-bit , a cryptographic hash function used by AURORA to generate deterministic topology hashes of CAD geometry for change detection and BOM matching.
SHAP	SHapley Additive exPlanations , a game-theory-based method for explaining ML model predictions; AURORA uses SHAP to provide human-readable explanations for CAD-BOM match scores.
SME	Subject Matter Expert , naval engineers and programme managers engaged during pilot validation and compliance rule catalogue development.
STEP	Standard for the Exchange of Product model data (ISO 10303) , the primary neutral CAD exchange format supported by AURORA's OCC-based parsing engine.
Surrogate Model	A lightweight ML model (here: PyTorch neural network) trained to approximate the outputs of an expensive computation (FEA/CFD), reducing evaluation time by ~95% during optimisation.
UAT	User Acceptance Testing , a validation phase where end-users test the system against real-world scenarios to confirm it meets operational requirements.
UUID	Universally Unique Identifier , a 128-bit identifier used by AURORA to uniquely tag every component in the Digital Twin and knowledge graph.
XGBoost	eXtreme Gradient Boosting , an ensemble ML algorithm used in AURORA's CAD-BOM Synchronisation Engine for multi-signal match confidence scoring.
XCAF	Extended CAD Application Framework , an OCC data model for managing multi-level CAD assembly hierarchies; used by AURORA during CAD parsing to extract component tree structure.
XML / CSV	eXtensible Markup Language / Comma-Separated Values , structured data formats used for BOM ingestion, schedule export, and PLM data interchange in AURORA.
Zero-Trust	A Phase 6 security architecture principle assuming no implicit trust between system components; enforced in AURORA via mTLS between microservices and strict RBAC.